



The
BRITISH UNIVERSITY
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Enterprise Network Design and Configuration for Global Widgets Inc.

NETWORK MANAGEMENT AND OPERATION PROJECT PROPOSAL

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ABSTRACT

Network infrastructure at the heart of seamless communication, security and operational efficiency in modern enterprises. This project involves the design and implementation of an enterprise-grade, scalable and extremely secure enterprise network for the overseas based company named Global Widgets Inc. To bring the basic network management features such as VLANs to separate the traffic, OSPF for Routing to find the least cost path to a destination, ACLs for security, and STP to prevent loops and allow redundancy. It will also utilize load balancing techniques to evenly distribute traffic on the network to prevent congestion and improve overall performance. Furthermore, this integration of automated network monitoring tools is also included in this project which will increase the efficiency, decrease the downtimes and field maintenance. The ability to monitor its key components in real-time, detect any anomalies, and issue predictive maintenance features to allow admin to act fast in advance of a potential failure. Automation frameworks like intent-based networking (IBN) will be delved into to help dynamically optimize network configurations to meet workload needs without human input. This project strives to design an enterprise network infrastructure that is impacted by the new trending networking technologies and the best practices around ensuring that the enterprise network infrastructure is resilient and future ready. This will primarily be centered around reliable high availability, hassle-free scalability, and providing high-level security layers as per the growing demands of the organization and compliance with industrial standards and statutory regulations.

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1 PROJECT DESCRIPTION

Enterprise networks enable firms to communicate, be safe, and be productive. This project intends to create a network that is scalable, high-performing, and secure.

The primary components of the network are VLANs for security, OSPF routing to locate the optimum data pathways, and Access Control Lists (ACLs) to enforce security. We will also employ the Spanning Tree Protocol (STP) to eliminate network loops.

To make the network work better, we will use load balancing and methods to improve network performance, we will deploy load balancing and congestion-reduction technologies. We will also implement automatic monitoring to reduce downtime. Real-time monitoring will allow us to identify problems before they occur, and we will employ tools such as Intent-Based Networking (IBN) to automatically modify network settings.

Our objective is to build a robust, future-ready enterprise network that complies with legal requirements and industry best practices.

2 NETWORK DESIGN COMPONENTS:

2.1 WIDE AREA NETWORK TOPOLOGY AND CONNECTIVITY

For enabling smooth communication between different office locations, it is essential to have an optimized Wide Area Network (WAN). Optimized WAN connections must be used to interconnect different locations using technologies like Multiprotocol Label Switching (MPLS), Software-Defined WAN (SD-WAN), or Virtual Private Network (VPN) over broadband, so that secure and efficient connectivity is established.

Every site should have a highly organized Local Area Network (LAN) with the right subnetting mechanisms for efficient allocation of IP addresses and better network performance. The following Subnetting Strategy Can Be Followed:

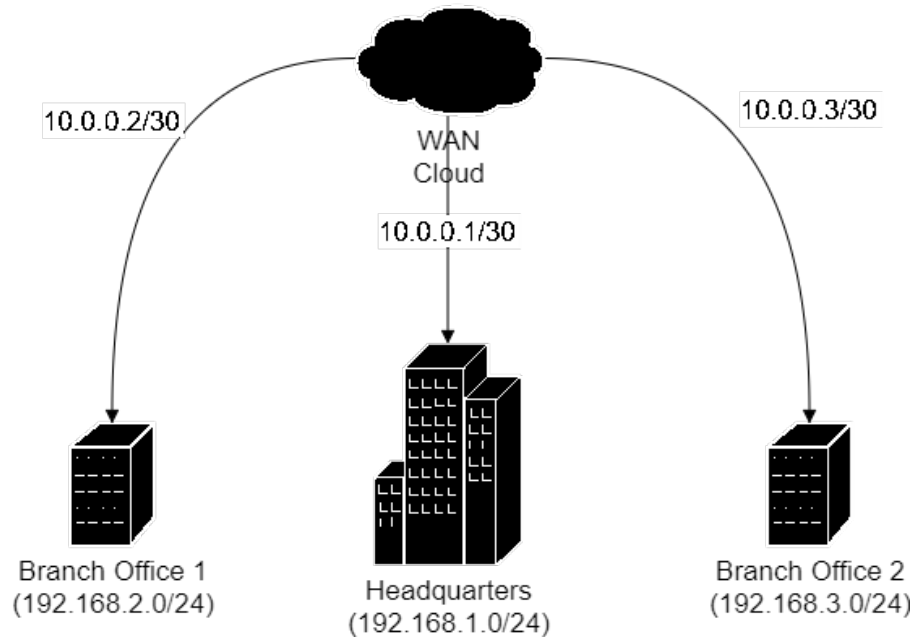


Figure 1 Topology & Wan Connectivity

- **Headquarters (HQ):** 192.168.1.0/24
- **Branch Office 1:** 192.168.2.0/24
- **Branch Office 2:** 192.168.3.0/24
- **WAN Link:** 10.0.0.0/30 (point-to-point connections)

2.2 VLAN CONFIGURATION:

To help increase security and optimize network performance, it is necessary that VLANs are created to logically divide network traffic. VLANs will divide network traffic based on function or department, reducing broadcast domains and increasing security. VLAN IDs should be assigned as follows:

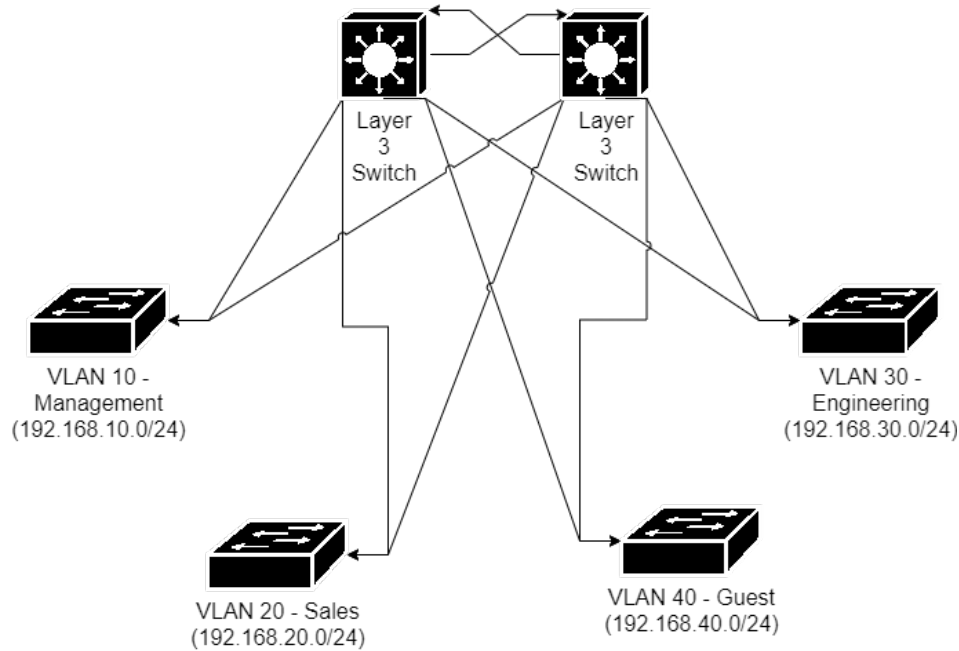


Figure 2 V-LAN Configuration

- **VLAN 10:** Management (192.168.10.0/24)
- **VLAN 20:** Sales (192.168.20.0/24)
- **VLAN 30:** Engineering (192.168.30.0/24)
- **VLAN 40:** Guest (192.168.40.0/24)

Inter-VLAN routing should be implemented using either a router or a Layer 3 switch to enable the necessary inter-VLAN communication. In addition to this, VLAN trunks should also be set up across switches using the IEEE 802.1Q standard to ensure inter-site inter-VLAN communication.

2.3 OSPF ROUTING PROTOCOL:

Open Shortest Path First (OSPF) has been chosen as the major routing protocol because of the ease with which it can adapt to changes within the network. OSPF has the following basic configurations:

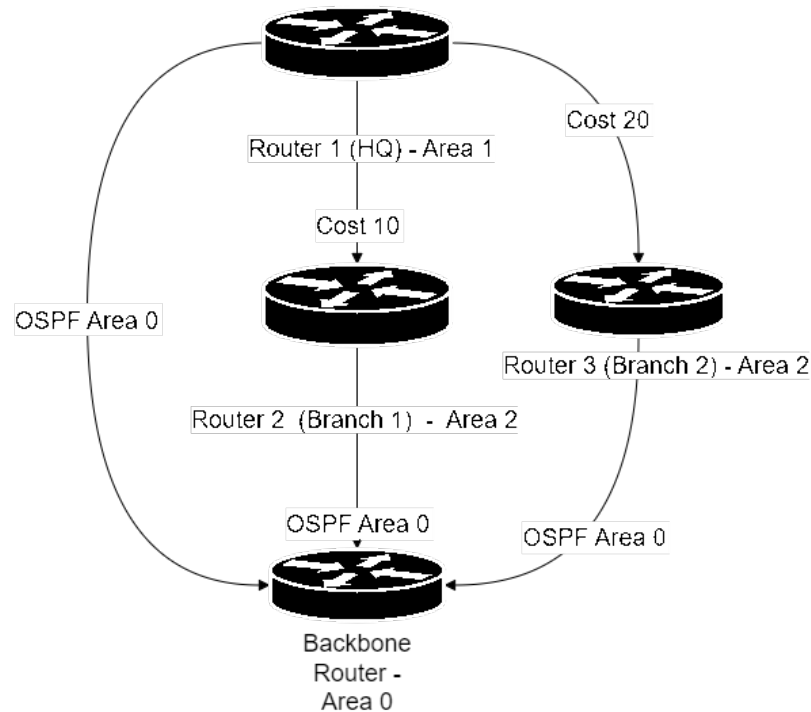


Figure 3 OSPF Routing

- **Assign OSPF Areas:**
 - Area 0 (Backbone) for inter-site connectivity
 - Area 1 for HQ LAN
 - Area 2 for Branch Offices
- **Optimize Cost Metrics:** Assign link cost values to optimize path selection.
- **Fast Convergence:** Enable OSPF hello and dead interval tuning to ensure quick recovery from failures.
- **Authentication:** Implement OSPF authentication to prevent unauthorized routing updates.

2.4 ACCESS CONTROL LISTS (ACLs):

Access Control Lists (ACLs) will be implemented to provide additional security to the network by controlling traffic flow and preventing unauthorized access. Sample ACL rules are

- **Permit Internal Users:** Allow LAN users to access the internet.

```
access-list 100 permit ip 192.168.0.0 0.0.255.255 any
```

- **Block Unauthorized Access to Servers:** Prevent access to internal servers from guest VLAN.

```
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
```

- **Allow Specific Traffic:** Allow only HTTP and HTTPS traffic to external web servers.

- ```
access-list 102 permit tcp any any eq 80
```

```
access-list 102 permit tcp any any eq 443
```

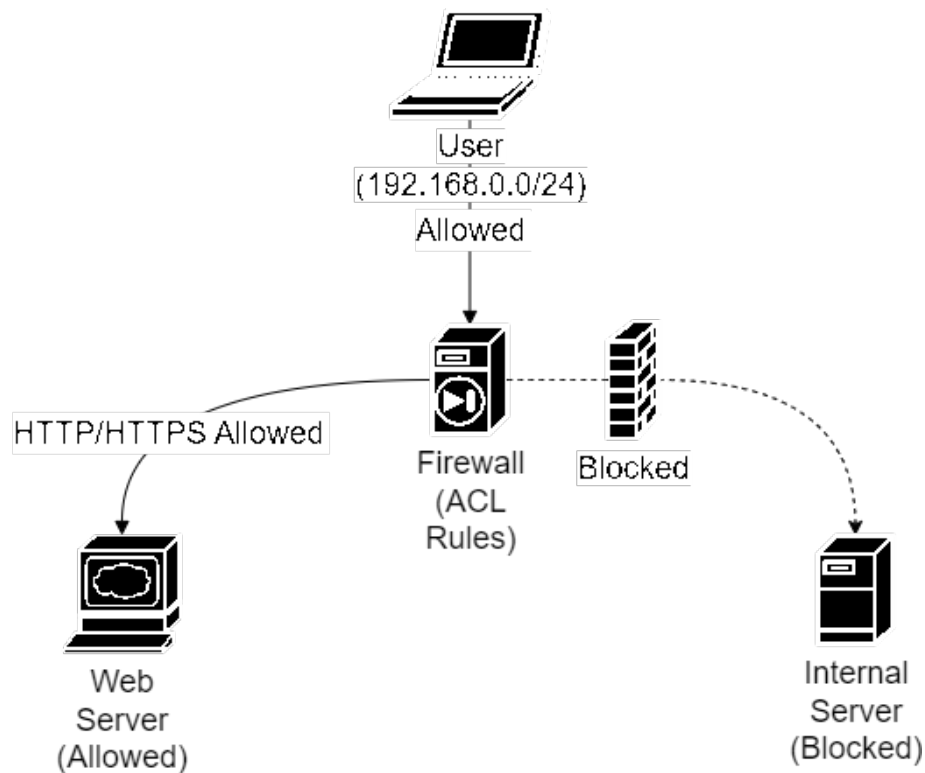


Figure 4 Access Control Lists (ACLs)

## 3 METHODOLOGY

---

This section describes the step-by-step implementation of the proposed enterprise network within the Cisco Packet Tracer simulation environment. The procedures are organized into subsections covering simulation setup, topology construction, device configuration, protocol deployment, and validation.

### 3.1 SIMULATION ENVIRONMENT

The network was modeled and tested in Cisco Packet Tracer the simulation utilized the following devices models:

- Cisco 2811 routers for HQ and branch WAN connectivity
- Cisco 2950-24 switches for each branch site
- Cisco 3560-24PS multilayer switch for the HQ
- End-user PCs representing Sales, Engineering, Management, and Guest workgroups

## 3.2 TOPOLOGY

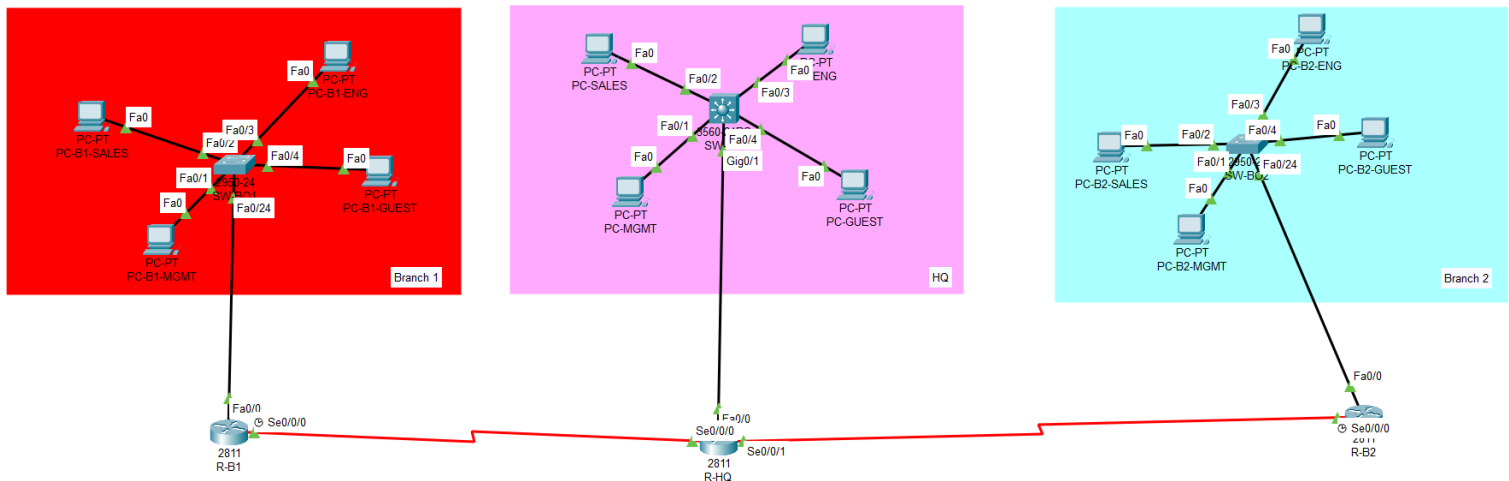


Figure 5 Cisco Packet Tracer Network

Figure 5 shows the simplified implementation of the three-site star network topology, where each branch and the headquarters are connected using point-to-point serial links, using the 10.0.0.0/30 sub-network for inter-site communication. Routers were placed, and Fast Ethernet connections were made to local switches. End devices were then attached to switch access ports corresponding to departmental VLANs.

## 3.3 INTERFACE AND IP ADDRESS CONFIGURATION

For each router and switch, the following interface configurations were applied:

- Router FastEthernet interfaces (Fa0/0) were assigned to local LAN subnets (e.g., HQ: 10.0.0.1/24; Branch-1: 10.0.0.2/24; Branch-2: 10.0.0.3/24).
- Serial interfaces (Se0/0/0 and Se0/0/1) were configured with DCE/DTE settings, a clock rate of 1.544 Mbps on the DCE end, and IP addressing from the 10.0.0.0/30 pool.

- Switch ports Fa0/1 through Fa0/4 were designated based on VLAN assignments.

### 3.4 VLAN DEPLOYMENT AND INTER-VLAN ROUTING

VLANs were created on each switch to segment traffic logically:

- VLAN 10: Management (192.168.10.0/24)
- VLAN 20: Sales (192.168.20.0/24)
- VLAN 30: Engineering (192.168.30.0/24)
- VLAN 40: Guest (192.168.40.0/24)

#### 3.4.1 IP Scheme per Site and VLAN:

| VLAN                   | Subnet          | Router GW IP | Example PC IPs     |
|------------------------|-----------------|--------------|--------------------|
| <b>VLAN 10 (MGMT)</b>  | 192.168.10.0/24 | 192.168.10.1 | 192.168.10.10, .11 |
| <b>VLAN 20 (SALES)</b> | 192.168.20.0/24 | 192.168.20.1 | 192.168.20.10, .11 |
| <b>VLAN 30 (ENG)</b>   | 192.168.30.0/24 | 192.168.30.1 | 192.168.30.10, .11 |
| <b>VLAN 40 (GUEST)</b> | 192.168.40.0/24 | 192.168.40.1 | 192.168.40.10, .11 |

*Table 1 HQ (pink area)*

| VLAN                   | Subnet          | Router GW IP | Example PC IPs     |
|------------------------|-----------------|--------------|--------------------|
| <b>VLAN 10 (MGMT)</b>  | 192.168.11.0/24 | 192.168.11.1 | 192.168.11.10, .11 |
| <b>VLAN 20 (SALES)</b> | 192.168.21.0/24 | 192.168.21.1 | 192.168.21.10, .11 |
| <b>VLAN 30 (ENG)</b>   | 192.168.31.0/24 | 192.168.31.1 | 192.168.31.10, .11 |
| <b>VLAN 40 (GUEST)</b> | 192.168.41.0/24 | 192.168.41.1 | 192.168.41.10, .11 |

*Table 2 Branch 1 (red area)*

| VLAN                   | Subnet          | Router GW IP | Example PC IPs     |
|------------------------|-----------------|--------------|--------------------|
| <b>VLAN 10 (MGMT)</b>  | 192.168.12.0/24 | 192.168.12.1 | 192.168.12.10, .11 |
| <b>VLAN 20 (SALES)</b> | 192.168.22.0/24 | 192.168.22.1 | 192.168.22.10, .11 |
| <b>VLAN 30 (ENG)</b>   | 192.168.32.0/24 | 192.168.32.1 | 192.168.32.10, .11 |
| <b>VLAN 40 (GUEST)</b> | 192.168.42.0/24 | 192.168.42.1 | 192.168.42.10, .11 |

*Table 3 Branch 2 (blue area)*

| Link                 | HQ IP    | Branch IP | Subnet      |
|----------------------|----------|-----------|-------------|
| <b>HQ ↔ Branch 1</b> | 10.0.0.1 | 10.0.0.2  | 10.0.0.0/30 |
| <b>HQ ↔ Branch 2</b> | 10.0.0.5 | 10.0.0.6  | 10.0.0.4/30 |

*Table 4 WAN / Router-to-Router IPs*

| PC Name         | VLAN    | IP Address    | Subnet Mask   | Gateway      |
|-----------------|---------|---------------|---------------|--------------|
| <b>PC-MGMT</b>  | VLAN 10 | 192.168.10.10 | 255.255.255.0 | 192.168.10.1 |
| <b>PC-SALES</b> | VLAN 20 | 192.168.20.10 | 255.255.255.0 | 192.168.20.1 |
| <b>PC-ENG</b>   | VLAN 30 | 192.168.30.10 | 255.255.255.0 | 192.168.30.1 |
| <b>PC-GUEST</b> | VLAN 40 | 192.168.40.10 | 255.255.255.0 | 192.168.40.1 |

*Table 5 HQ PCs (pink area)*

| PC Name            | VLAN    | IP Address    | Subnet Mask   | Gateway      |
|--------------------|---------|---------------|---------------|--------------|
| <b>PC-B1-MGMT</b>  | VLAN 10 | 192.168.11.10 | 255.255.255.0 | 192.168.11.1 |
| <b>PC-B1-SALES</b> | VLAN 20 | 192.168.21.10 | 255.255.255.0 | 192.168.21.1 |
| <b>PC-B1-ENG</b>   | VLAN 30 | 192.168.31.10 | 255.255.255.0 | 192.168.31.1 |
| <b>PC-B1-GUEST</b> | VLAN 40 | 192.168.41.10 | 255.255.255.0 | 192.168.41.1 |

*Table 6 Branch 1 PCs (red area)*

| PC Name            | VLAN    | IP Address    | Subnet Mask   | Gateway      |
|--------------------|---------|---------------|---------------|--------------|
| <b>PC-B2-MGMT</b>  | VLAN 10 | 192.168.12.10 | 255.255.255.0 | 192.168.12.1 |
| <b>PC-B2-SALES</b> | VLAN 20 | 192.168.22.10 | 255.255.255.0 | 192.168.22.1 |
| <b>PC-B2-ENG</b>   | VLAN 30 | 192.168.32.10 | 255.255.255.0 | 192.168.32.1 |
| <b>PC-B2-GUEST</b> | VLAN 40 | 192.168.42.10 | 255.255.255.0 | 192.168.42.1 |

*Table 7 Branch 2 PCs (blue area)*

### **3.5 OSPF ROUTING CONFIGURATION**

Open Shortest Path First (OSPF) was deployed to provide dynamic routing across all routers. The following process was executed:

1. Initialized OSPF process 1 on each router.
2. Advertised serial and LAN networks under appropriate areas (Area 0 for WAN, Area 1 for HQ LAN, Area 2 for branch LANs).
3. Tuned hello and dead intervals to 10 seconds and 40 seconds, respectively, for faster convergence.
4. Enabled MD5 authentication on all OSPF interfaces to secure routing updates.

### 3.6 ACCESS CONTROL AND SPANNING TREE ACCESS CONTROL LISTS

(ACLs) were applied on router interfaces to enforce security policies:

- ACL 100 permitted internal LAN hosts Internet access.
- ACL 101 denied Guest VLAN traffic to the Management subnet.
- ACL 102 permitted only HTTP and HTTPS outbound from internal users.

```
Switch>en
Switch#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Management
Switch(config-vlan)#vlan 20
Switch(config-vlan)#Sales
Switch(config-vlan)#^
% Invalid input detected at '^' marker.

Switch(config-vlan)#name Sales
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name Engineering
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name Guest
Switch(config-vlan)#int
Switch(config-vlan)#ex
Switch(config)#int
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fa0/3
Switch(config-if)#switchport access vlan 30
Switch(config-if)#ex
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#ex
Switch(config)#interface fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#ex
Switch(config)#int
Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switc

Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switc
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk encap
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#ex
Switch(config)#ex
Switch#
%SYS-S-CONFIG_I: Configured from console by console

Switch#write mem
Switch#write memory
Building configuration...
[OK]
Switch#
Switch#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#
?Bad filename
%Error parsing filename (Bad file number)
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/2
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/4
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/5
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/24
Switch(config-if)#
Switch(config-if)#exit
```

*Figure 6 ACL CLI configuration*



### 3.7 VALIDATION AND TESTING

Connectivity and protocol operation were validated through a series of tests:

- Ping tests between PCs across VLANs and sites to confirm inter-VLAN and inter-site reachability.

PC- B1 sales pinging PC HQ and PC-b2 (192.168.21.10 ping 192.168.20.10 and 192.168.22.10)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=17ms TTL=126
Reply from 192.168.20.10: bytes=32 time=17ms TTL=126
Reply from 192.168.20.10: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.20.10:
 Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
 Minimum = 11ms, Maximum = 17ms, Average = 15ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=19ms TTL=126
Reply from 192.168.20.10: bytes=32 time=8ms TTL=126
Reply from 192.168.20.10: bytes=32 time=10ms TTL=126
Reply from 192.168.20.10: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.20.10:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
 Minimum = 8ms, Maximum = 19ms, Average = 11ms

C:\>ping 192.168.22.10

Pinging 192.168.22.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.22.10: bytes=32 time=2ms TTL=125
Reply from 192.168.22.10: bytes=32 time=21ms TTL=125
Reply from 192.168.22.10: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.22.10:
 Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
 Minimum = 2ms, Maximum = 21ms, Average = 8ms

C:\>ping 192.168.22.10

Pinging 192.168.22.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.22.10: bytes=32 time=2ms TTL=125
Reply from 192.168.22.10: bytes=32 time=21ms TTL=125
Reply from 192.168.22.10: bytes=32 time=20ms TTL=125
Reply from 192.168.22.10: bytes=32 time=21ms TTL=125

Ping statistics for 192.168.22.10:
 Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
 Minimum = 2ms, Maximum = 21ms, Average = 8ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=19ms TTL=126
Reply from 192.168.20.10: bytes=32 time=8ms TTL=126
Reply from 192.168.20.10: bytes=32 time=10ms TTL=126
Reply from 192.168.20.10: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.20.10:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
 Minimum = 8ms, Maximum = 19ms, Average = 11ms

C:\>ping 192.168.22.10

Pinging 192.168.22.10 with 32 bytes of data:

Reply from 192.168.22.10: bytes=32 time=21ms TTL=125
Reply from 192.168.22.10: bytes=32 time=2ms TTL=125
Reply from 192.168.22.10: bytes=32 time=20ms TTL=125
Reply from 192.168.22.10: bytes=32 time=21ms TTL=125

Ping statistics for 192.168.22.10:
 Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
 Minimum = 2ms, Maximum = 21ms, Average = 16ms
```

Figure 7 Example ping test

### 3.8 VERIFICATION

To the validity of the practical implementation:

- “show ip route” and “show ip ospf” neighbor commands to verify OSPF adjacency and routing tables.
- “show vlan brief” to confirm VLAN membership and trunk status.

CLI verification outputs for each router are presented in the flowing pages.

#### 3.8.1 HQ Router (R-HQ) Interface and VLAN Sub interfaces

```
R-HQ>en
R-HQ#show ip interface brief
```

| Interface          | IP-Address   | OK? | Method | Status                | Protocol |
|--------------------|--------------|-----|--------|-----------------------|----------|
| FastEthernet0/0    | unassigned   | YES | unset  | up                    | up       |
| FastEthernet0/0.10 | 192.168.10.1 | YES | manual | up                    | up       |
| FastEthernet0/0.20 | 192.168.20.1 | YES | manual | up                    | up       |
| FastEthernet0/0.30 | 192.168.30.1 | YES | manual | up                    | up       |
| FastEthernet0/0.40 | 192.168.40.1 | YES | manual | up                    | up       |
| FastEthernet0/1    | unassigned   | YES | unset  | administratively down | down     |
| Serial0/0/0        | 10.0.0.1     | YES | manual | up                    | up       |
| Serial0/0/1        | 10.0.0.5     | YES | manual | up                    | up       |
| Vlan1              | unassigned   | YES | unset  | administratively down | down     |

```
R-HQ#
R-HQ#show ip ospf neighbor
```

| Neighbor ID  | Pri | State   | Dead Time | Address  | Interface   |
|--------------|-----|---------|-----------|----------|-------------|
| 192.168.42.1 | 0   | FULL/ - | 00:00:32  | 10.0.0.6 | Serial0/0/1 |
| 192.168.41.1 | 0   | FULL/ - | 00:00:32  | 10.0.0.2 | Serial0/0/0 |

```
R-HQ#show ip route ospf
```

| O | 192.168.11.0 | [110/65] | via 10.0.0.2, | 00:02:16, | Serial0/0/0 |
|---|--------------|----------|---------------|-----------|-------------|
| O | 192.168.12.0 | [110/65] | via 10.0.0.6, | 00:02:16, | Serial0/0/1 |
| O | 192.168.21.0 | [110/65] | via 10.0.0.2, | 00:02:16, | Serial0/0/0 |
| O | 192.168.22.0 | [110/65] | via 10.0.0.6, | 00:02:16, | Serial0/0/1 |
| O | 192.168.31.0 | [110/65] | via 10.0.0.2, | 00:02:16, | Serial0/0/0 |
| O | 192.168.32.0 | [110/65] | via 10.0.0.6, | 00:02:16, | Serial0/0/1 |
| O | 192.168.41.0 | [110/65] | via 10.0.0.2, | 00:02:16, | Serial0/0/0 |
| O | 192.168.42.0 | [110/65] | via 10.0.0.6, | 00:02:16, | Serial0/0/1 |

```

VLAN Name Status Ports

1 default active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
R-HQ#

```

### 3.8.2 Branch-1 Router (R-B1) Serial and VLAN Interfaces

```

R-B1>en
R-B1#show ip interface brief

```

| Interface          | IP-Address   | OK? | Method | Status                | Protocol |
|--------------------|--------------|-----|--------|-----------------------|----------|
| FastEthernet0/0    | unassigned   | YES | unset  | up                    | up       |
| FastEthernet0/0.10 | 192.168.11.1 | YES | manual | up                    | up       |
| FastEthernet0/0.20 | 192.168.21.1 | YES | manual | up                    | up       |
| FastEthernet0/0.30 | 192.168.31.1 | YES | manual | up                    | up       |
| FastEthernet0/0.40 | 192.168.41.1 | YES | manual | up                    | up       |
| FastEthernet0/1    | unassigned   | YES | unset  | administratively down | down     |
| Serial0/0/0        | 10.0.0.2     | YES | manual | up                    | up       |
| Serial0/0/1        | unassigned   | YES | unset  | administratively down | down     |
| Vlan1              | unassigned   | YES | unset  | administratively down | down     |

```

R-B1#show ip ospf neighbor

```

| Neighbor ID  | Pri | State   | Dead Time | Address  | Interface   |
|--------------|-----|---------|-----------|----------|-------------|
| 192.168.40.1 | 0   | FULL/ - | 00:00:31  | 10.0.0.1 | Serial0/0/0 |

```

R-B1#show ip route ospf

```

10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks

| O | 10.0.0.4 [110/128]     | via 10.0.0.1, 00:04:32, Serial0/0/0 |
|---|------------------------|-------------------------------------|
| O | 192.168.10.0 [110/65]  | via 10.0.0.1, 00:04:32, Serial0/0/0 |
| O | 192.168.12.0 [110/129] | via 10.0.0.1, 00:04:22, Serial0/0/0 |
| O | 192.168.20.0 [110/65]  | via 10.0.0.1, 00:04:32, Serial0/0/0 |
| O | 192.168.22.0 [110/129] | via 10.0.0.1, 00:04:22, Serial0/0/0 |
| O | 192.168.30.0 [110/65]  | via 10.0.0.1, 00:04:32, Serial0/0/0 |
| O | 192.168.32.0 [110/129] | via 10.0.0.1, 00:04:22, Serial0/0/0 |
| O | 192.168.40.0 [110/65]  | via 10.0.0.1, 00:04:32, Serial0/0/0 |
| O | 192.168.42.0 [110/129] | via 10.0.0.1, 00:04:22, Serial0/0/0 |

### 3.8.3 Branch-2 Router (R-B2) Serial and VLAN Interfaces

```
R-B2>en
R-B2#show ip interface brief
```

| Interface          | IP-Address   | OK? | Method | Status                | Protocol |
|--------------------|--------------|-----|--------|-----------------------|----------|
| FastEthernet0/0    | unassigned   | YES | unset  | up                    | up       |
| FastEthernet0/0.10 | 192.168.12.1 | YES | manual | up                    | up       |
| FastEthernet0/0.20 | 192.168.22.1 | YES | manual | up                    | up       |
| FastEthernet0/0.30 | 192.168.32.1 | YES | manual | up                    | up       |
| FastEthernet0/0.40 | 192.168.42.1 | YES | manual | up                    | up       |
| FastEthernet0/1    | unassigned   | YES | unset  | administratively down | down     |
| Serial0/0/0        | 10.0.0.6     | YES | manual | up                    | up       |
| Serial0/0/1        | unassigned   | YES | unset  | administratively down | down     |
| Vlan1              | unassigned   | YES | unset  | administratively down | down     |

```
R-B2#show ip ospf neighbor
```

| Neighbor ID  | Pri | State   | Dead Time | Address  | Interface   |
|--------------|-----|---------|-----------|----------|-------------|
| 192.168.40.1 | 0   | FULL/ - | 00:00:30  | 10.0.0.5 | Serial0/0/0 |

```
R-B2#show ip route ospf
```

```

10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O 10.0.0.0 [110/128] via 10.0.0.5, 00:14:51, Serial0/0/0
O 192.168.10.0 [110/65] via 10.0.0.5, 00:14:51, Serial0/0/0
O 192.168.11.0 [110/129] via 10.0.0.5, 00:14:41, Serial0/0/0
O 192.168.20.0 [110/65] via 10.0.0.5, 00:14:51, Serial0/0/0
O 192.168.21.0 [110/129] via 10.0.0.5, 00:14:41, Serial0/0/0
O 192.168.30.0 [110/65] via 10.0.0.5, 00:14:51, Serial0/0/0
O 192.168.31.0 [110/129] via 10.0.0.5, 00:14:41, Serial0/0/0
O 192.168.40.0 [110/65] via 10.0.0.5, 00:14:51, Serial0/0/0
O 192.168.41.0 [110/129] via 10.0.0.5, 00:14:41, Serial0/0/0

```

## 4 LITERATURE REVIEW

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### 4.1 RESEARCH PAPERS:

#### 4.1.1 Paper 1: "An Abstraction-Driven Approach to Systematic Enterprise Network Design, Configuration, and Validation" by Yu-Wei Sung

This paper presents a straightforward approach to make enterprise network design and configuration easier. Instead of focusing on specific hardware setups, this method prioritizes high-level representations to improve the management of VLANs and ACLs. The paper highlights these key points:

- **Automation:** The abstraction-based approach helps network administrators test configurations before deployment. This process reduces human errors and improves operational efficiency.
- **VLAN Segmentation:** This feature keeps different departments in an organization separate, improving security and performance.
- **Policy Adherence:** Using abstraction makes network setups more organized. This helps ensure security rules are followed and allows for easy large-scale deployments.
- **Security and Policy Control:** We include automated checks to reduce risks from manual setup mistakes, ensuring consistent enforcement of security policies.
- **Scalability:** Abstraction helps expand the network easily by automating repetitive tasks and adjusting to changing workload needs.

**4.1.2 Paper 2: "Towards Systematic Design of Enterprise Networks" by Yu-Wei Eric Sung, Sanjay G. Rao, Geoffrey G. Xie, and David A. Maltz**

This study takes a manual and organized approach to network architecture. It focuses in VLAN segmentation, routing protocols, and security policy enforcement. The study provides a framework for ensuring network regulations meet business requirements while also being scalable and safe.

- **Manual Systematic Approach:** This method uses predefined plans to create a structured design process. However, it requires a strong understanding of best practices.
- **Reachability Control:** The framework ensures secure communication between departments and limits access to unauthorized users.
- **Real-World Case Studies:** The paper includes real examples that show the negative effects of poor network segmentation and misaligned policies.
- **Security Focus:** This approach uses step-by-step plans to enforce security policies and maintain compliance.
- **Policy-Driven Stability:** The framework is meant for organizations that want long-term network stability rather than frequent changes, making it suitable for static enterprise environments.

## 4.2 COMPARISON & INSIGHTS:

| Feature            | Abstraction-Driven Approach                             | Systematic Manual Approach                                    |
|--------------------|---------------------------------------------------------|---------------------------------------------------------------|
| <b>Automation</b>  | Fully automated with configuration validation           | Requires manual configuration with structured execution plans |
| <b>Security</b>    | Automated policy validation and enforcement             | Relies on predefined security execution plans                 |
| <b>Scalability</b> | Suitable for dynamic environments with frequent changes | Best for stable, policy-driven networks                       |
| <b>Complexity</b>  | Simplifies deployment and reduces human error           | Requires extensive knowledge and manual implementation        |
| <b>Flexibility</b> | Adapts to workload changes dynamically                  | Focuses on long-term stability with controlled modifications  |

*Table 8 Research comparison and insights*

### 4.2.1 Automation vs. Manual:

The first paper supports an approach for automating the tasks needed for design, facilitating the operation of large-scale enterprise networks, while the second paper relies on a structured manual process needing extensive knowledge of best practices in policy enforcement.

### 4.2.2 Security & Policy Control:

Although both approaches consider security, the first paper validates security policies, automatically abstracting away the risks of manual planning and misconfigurations, while the second paper approach uses a sequence of deployable execution plans to keep the systems compliant.

### 4.2.3 Scalability & Flexibility:

Because the abstraction-based approach, automates many repetitive tasks and makes every process, scalable, it is a great fit for dynamic enterprise environments. The systematic design approach, in contrast, is more appropriate for stable, static, and carefully constructed networks that require stringent control.

#### **4.2.4 Nature of Use:**

An abstraction-driven approach works best for organizations having rapidly changing networking requirements, and a systematic design approach works well for enterprises that want to have a policy-driven security and stability.

#### **4.2.5 Findings:**

Both approaches seek to improve enterprise network design. The abstraction-driven approach is ideal for businesses that require flexibility and automation. On the other hand, the systematic manual technique is preferable for businesses that prioritize security and an organized way of working execution.

## **5 CONCLUSION:**

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This proposal outlines a comprehensive plan for designing and building a global enterprise network for GlobalWidgets Inc. Using the best practices of VLAN segmentation, OSPF routing, ACL process, and STP configuration, the completion of the project aims to converge VLAN into high-performance, secure, and expandable network infrastructure and minimize the risk of risk. Furthermore, design methodology will also be augmented with academic research insights to enable an efficient and effectively organized enterprise network. In addition, this approach will bring in flexibility and quick adaptability to the changing requirements of enterprises by incorporating SDN principles and network automation. With solid planning, implementation, and monitoring processes, this enterprise network will serve you to run at the best capacity and grow the business.



## **REFERENCES:**

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- [1]. Yu-Wei Sung. "An Abstraction-Driven Approach to Systematic Enterprise Network Design, Configuration, and Validation."**
- [2]. Yu-Wei Eric Sung, Sanjay G. Rao, Geoffrey G. Xie, David A. Maltz. "Towards Systematic Design of Enterprise Networks."**